

LLM4SE - Sample Questions 2025

Politecnico di Torino

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Collected Questions from LLM & SE Sample Sets

Part 1: Large Language Models (LLM)

Question 1 (*Parameter Counting - Decoder Only*)

For the following **decoder-only** transformer model, compute:

1. The number of parameters for the embedding layer.
2. The number of parameters of the transformer (excluding the embedding layer, or head layer(s)).

Note: assume that none of the layers uses a bias term.

Hyperparameters:

- Vocabulary size $V = 30,000$
- Embedding/model dimension $d = 768$
- Positional embeddings: learned, for 512 positions
- Number of layers $L = 12$
- Number of attention heads $H = 12$
- FF-NN projection dimension: 3072
- Head dimension $d_k = d_v = 64$

Question 2 (*PEFT Techniques*)

Which of the following is **NOT** a Parameter-Efficient Fine-Tuning (PEFT) technique?

- a) Dynamic quantization
- b) Adapter layers
- c) Prompt tuning
- d) LoRA

Question 3 (Attention Computation)

The following vectors for the input tokens are given:

Inputs: $[2, 0]$, $[0, 2]$

The following matrices W_q, W_k, W_v are given:

$$W_q = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}, \quad W_k = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 0 & 1 \end{bmatrix}, \quad W_v = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Compute Attention $(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$ for the **first token**.

Question 4 (Model Architecture)

Which one of these models is a **decoder-only**?

- a) BERT
 - b) GPT-2
 - c) T5
 - d) All options are decoder-only models
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Question 5 (Causal Language Modeling)

In Causal Language Modeling, the word "causal" means that:

- a) The output is generated considering only the current token.
 - b) The output is generated considering only the previous tokens.
 - c) The output is generated considering only the previous and current tokens.
 - d) Only the decoder is used during inference.
 - e) The attention mechanism is adopted.
 - f) The surrounding context must also be used for token generation.
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Question 6 (BLEU Score)

- A model generates the sentence: **"the fast brown fox jumped the bored dog"**
- The target sentence is: **"the quick brown fox jumps over the lazy dog"**

Compute the **BLEU-2** score for this generation.

Remember:

$$BLEU-n = BP \cdot \exp\left(\frac{1}{n} \sum_i \log(\text{precision}_i)\right)$$

$$\text{Brevity Penalty } BP = \begin{cases} 1 & \text{if } g > r \\ e^{(1-\frac{r}{g})} & \text{if } g \leq r \end{cases}$$

Where precision is the precision for the specific i-gram, g and r are the lengths of the generated and reference texts, respectively. Consider each word to be a token.

Question 7 (Quantization)

What is the primary difference between **static** and **dynamic** quantization?

- a) Dynamic quantization computes scaling values for each activation during inference.
 - b) Dynamic quantization applies to weights only, while static applies to weights and activations.
 - c) Static quantization is slower but less accurate than dynamic quantization.
 - d) Dynamic Quantization requires a calibration step before the quantization.
 - e) None of the others.
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Question 8 (BPE Tokenization)

You are given the following corpus of text: **abababab**

Apply BPE to extract **3 tokens** (excluding 'a' and 'b'). Write the tokens and their extraction frequency (i.e., the frequency computed at the step when they were merged). *Note: when doing substitution, work from left to right.*

Question 9 (LoRA)

In LoRA, what does the process assume about the weight update matrix?

- a) That it is low rank or is near low rank.
 - b) That it replaces the initial weight matrix.
 - c) That it is initialized with numbers taken from a normal distribution $\mathcal{N}(0, 1)$.
 - d) That it is fixed (or frozen) throughout training.
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Question 10 (Sampling)

Which one of these is a **deterministic** sampling approach? Select all correct answers.

- a) Temperature sampling with $T = 0$
 - b) Temperature sampling, with $0 < T < 1$
 - c) Temperature sampling, with $T > 1$
 - d) Beam Search
 - e) Greedy sampling
 - f) Top-k sampling
 - g) Top-p sampling
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Part 2: Software Engineering (SE)

Question 11 (Metrics)

Consider the following cosine similarity table. Which of the following threshold is more dependable in terms of TPR and FPR?

$$T = 0.50, 0.75, 0.95$$

	Exp1	Exp2	Exp3	Exp4
Act1	0,86	0,84	0,55	0,42
Act2	0,72	0,92	0,12	0,53
Act3	0,81	0,16	0,78	0,55
Act4	0,44	0,75	0,33	0,93

Question 12 (Testing & Mutation)

Consider the following function, proposed mutants and generated tests. Compute the **mutation score** of the test suite.

```
1 def calculate_final_price(base_price, discount, tax_rate, membership_status,
2   loyalty_points):
3     discounted_price = base_price * (1 - discount/100)
4     loyalty_discount = loyalty_points * 0.05
5
6     discounted_price -= loyalty_discount
7
8     final_price_with_tax = discounted_price * (1 + tax_rate / 100)
9
10    if membership_status:
11        final_price_with_tax *= 0.9
12
13    if final_price_with_tax < 0:
14        final_price_with_tax = 0
15
16    return final_price_with_tax
```

Mutants:

- `discounted_price = base_price * (1 + discount/100)`
- `loyalty_discount = loyalty_points * 0.1`
- `if not membership_status:`

Tests:

- T1: `assert calculate_final_price(100, 20, 10, True, 10) == 72.0`
- T2: `assert calculate_final_price(200, 10, 0, False, 0) == 180.0`
- T3: `assert calculate_final_price(100, 0, 0, False, 0) == 100.0`

Question 13 (*Few-Shot Prompting*)

Consider the following requirements. Write a **few-shot prompt** to extract OOP Classes for a software architecture design.

REQS: "The platform allows customers to browse menus from various restaurants, place food orders, and track the delivery in real-time. It also includes features for reviewing and rating restaurants, as well as providing personalized recommendations based on the customer's past orders. Additionally, platform managers can update restaurant menus, track orders, and manage customer feedback to ensure a smooth experience."

Question 14 (*Prompt Engineering - Router*)

Define a **Router prompt** for the LLAMA engine for the generation of the three parts of user stories based on the requirements (REQS) from the previous question. The router prompt must have a default branch.

Question 15 (*Evaluation*)

Which of the following is **not** a way to evaluate the correctness in generating code?

- a) functional correctness
 - b) execution time
 - c) execution success rate
 - d) blind peer reviews
 - e) developer feedback
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Question 16 (*Agentic Architecture*)

In a **Router Chain**, what is the role of the 'Default Chain'?

- A. To act as a fallback when no other subchain is selected.
 - B. To store previous results for future processing.
 - C. To route inputs to multiple subchains simultaneously.
 - D. To ensure inputs are processed in a sequential order.
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Question 17 (*Agentic Architecture*)

What is the main function of the **Reflection module** in an agentic architecture?

- A. To evaluate past decisions and improve future performance.
- B. To manage and store intermediate outputs in short-term memory.
- C. To dynamically route inputs to specialized subchains.
- D. To decompose tasks into smaller, sequential subgoals.

- E. To plan the most efficient strategy for task execution.
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Question 18 (*Prompting Strategy*)

Few-Shot Prompting is particularly effective in which of the following scenarios?

- A. When the task requires the model to autonomously explore various paths to a solution.
 - B. When there is sufficient prior training data, making examples redundant.
 - C. When the task demands clarity through structured reasoning without specific examples.
 - D. When a complex task needs to be demonstrated with multiple examples to guide model behavior.
 - E. When the output must follow a consistent tabular format for easy comprehension.
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Question 19 (*Datasets*)

What is a potential limitation when using **industrial datasets** for training LLMs in software engineering (LLM4SE)?

- A. They are often publicly available and lack restrictions on data usage.
 - B. They tend to focus primarily on high-level software concepts, neglecting specific code-based insights.
 - C. They contain sensitive and proprietary data, which could pose privacy and security concerns in the model's training.
 - D. They are mainly composed of graphical user interface (GUI) representations, making them less suitable for training on code-based tasks.
 - E. They primarily include bugs and patches but lack sufficient context on software development practices.
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